

could such a complex mechanism be reduced to a “neat” solution like the one proposed by McCarthy or Simon? The idea of boiling down “logic” into a semantic network didn’t fit in with the idea of the human brain being unimaginably complex.

Instead, Schank found himself agreeing with the approach employed by cognitive scientist Marvin Minsky of the Massachusetts Institute of Technology. Minsky and his team were the original “hackers” – it was in their lab that the term was born. Their idea was that each task required a specific solution and there’s no “one ring to rule them all” universal principle for artificial intelligence. Minsky’s team used to work on each problem individually, tweaking code – or “hacking away” – till they got it to work right. Schank described this philosophy towards AI as “scruffy”, in stark contrast to the “neat” solutions.

The result of the scruffy approach was that it made more rapid strides in the non-problem-solving aspects of AI than neat solutions. For example, MIT computer scientist Joseph Weizenbaum created a computer program called ELIZA, which has often been described as the first truly functional natural language processing automations – or rather, “chatbot”. Acting as a psychotherapist, it gave remarkably human-like answers to questions. For instance, if the user typed “My mother hates me”, ELIZA would respond with “Who else hates you?”

A lot of modern robotics use scruffy AI rather than neat AI because of the way the robots are meant to behave. A long sequence of logical formalisations would lead to more resource-intensive behaviour, which would in turn be slower. If the robots were instead programmed to perform specific tasks based on reacting to a certain stimulus – which is essentially the scruffy approach – they would function in a more practical manner.

Embodied and Computational Intelligence

Every single school of thought regarding AI was obsessed with the human brain and eventually, they all started hitting dead ends. In 1986, Rodney Brooks – a professor of robotics at MIT – wrote a monumental paper called *Elephants Don’t Play Chess*, in which he stated that reasoning about the physical world is much simpler than interacting with it. He started focusing his work in AI on creating biologically-inspired robots.

While this is seen as the point where computational intelligence took off, its roots can be traced back a few more years to the emergence